Smart Protein Project
NIFTEM-K

NIFTEM KUNDLI
ज्ञान · नवोन्मेष · विस्तार

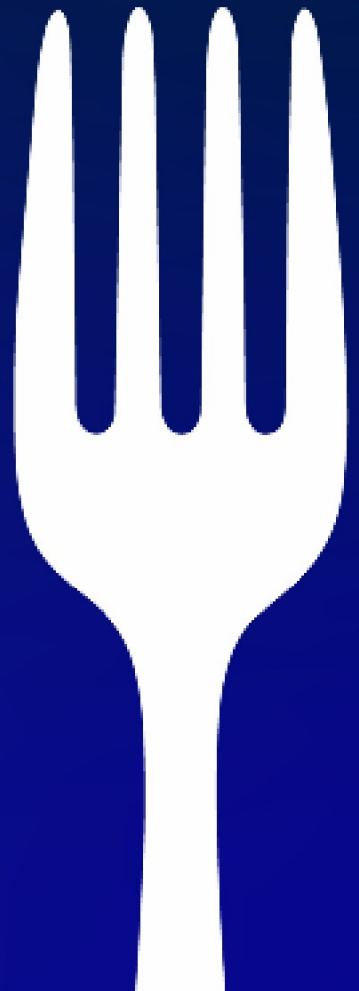
MOFPI
MINISTRY OF FOOD PROCESSING INDUSTRIES
GOVERNMENT OF INDIA

hackthefork.in

HACK THE FORK 2026 INDIA EDITION

Upcycle Compass

One by-product batch, one protein target: Go, No-go or Test-first, and what to test next



TEAM [Team name]

PROBLEM STATEMENT 02 · Upcycling India's By-products for Alternative Proteins

MEMBERS [Member names]

The Problem

A mill owner or plant-protein R&D head cannot quickly tell whether **one oilseed-cake batch** is safe for food, what processing it needs, what a kg of its protein would cost against imported pea isolate, and what to test first.

~90%

of plant-protein ingredients used in India are imported

PBFIA via Agron Food Processing · published

47%

of 118 Indian groundnut-cake samples met ≤ 20 ppb aflatoxin B1 (feed grade)

Murugeswari 2023 · published

₹518

per kg protein: modelled landed cost of imported pea isolate (range ₹431 to ₹619)

Our pea scenario · assumed, grade D

Why it hasn't been solved

- AI food platforms chase novel molecules, not commodity by-products
- Safety, processing and cost are studied separately, often per kg product, not per kg protein
- Lot data is scarce: most numbers are literature ranges, not this batch

Impact if solved

- Mills can offer cake to food buyers, not only as feed
- Brands get a local protein option priced per kg protein against pea
- Labs test in the right order: safety first, failing lots diverted early

Our Solution

1 What it does

- **In:** one batch (lab values, lot records) + one protein target
- **Out:** GO / NO-GO / TEST-FIRST, with safety, technical, cost and evidence shown separately
- Plus candidate routes, ₹/kg protein vs imported pea, and the test to run next

2 Target customers

- Oil mills selling cake at feed prices
- R&D heads at Indian plant-protein brands hedging pea imports
- Labs, institutes and incubators choosing which pilots to fund

3 Scale-up in the next 6 months

- **Month 0–1:** safety assays on 3–5 food-grade lots
- **Month 1–4:** bench trials of 2 routes; sensory and functional tests
- **Month 4–6:** ≈10 t pilot run; re-run on measured data; customer samples

4 Resources we need

- A food-grade oil mill and an accredited (NABL) lab
- A brand willing to test samples, and a pilot-plant slot
- Budget set from lab quotes (the tool will not guess it)

The Model

Batch + target
lab values, lot records, protein target

→ **Safety gates**
G1–G7, measured evidence only

→ **Diagnosis**
what blocks this use

→ **Candidate routes**
extrusion, phytase, alkaline extraction

→ **Mass balance + cost**
₹/kg protein vs landed pea

→ **Decision**
verdict, readiness, next test, 6-month plan

Data used

- 36 cited sources; 75 research records; 8 functional properties
- Every value tagged measured, published, assumed or synthetic, with grade and source
- Inputs as ranges, sampled 8,192 times (Sobol quasi-Monte Carlo)

AI and what makes it different

- A probabilistic decision engine, not a chatbot
- The LLM (Amazon Bedrock) only parses text and explains; a verifier rejects any number not in the engine output
- Safety first, priced per kg protein, and says what it does not know

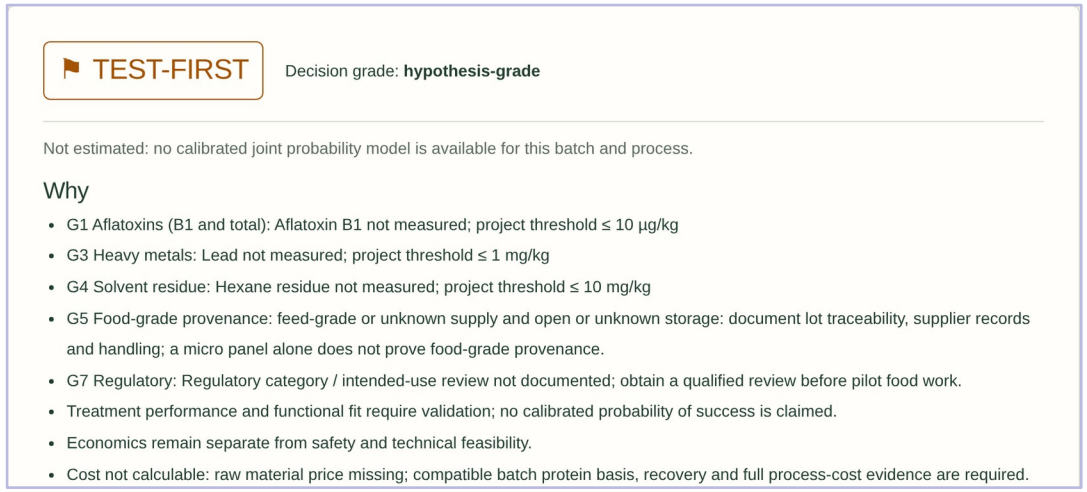
Results

- 5 presets, each analysed in about 0.2 s
- 170 automated tests pass
- Offline: 10 stamped snapshots, English and Hindi
- One lab value changes the verdict (lead 1.6 mg/kg → NO-GO)

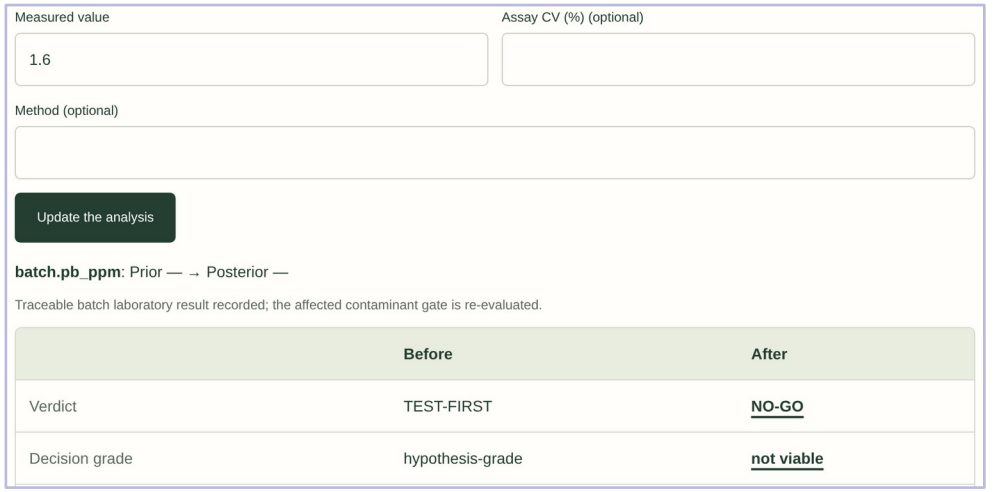
Built with Python, NumPy, SciPy, FastAPI, Pydantic, React + TypeScript and SQLite; Amazon Bedrock as the optional LLM, with a fixed template as fallback.

Prototype Demo

1 Sunflower → Protein Rich Flour: TEST-FIRST



2 One lab result (lead 1.6 mg/kg) turns it into NO-GO



Run a preset

7 contaminant checks unmeasured, so no bars are drawn and nothing is invented

Read the evidence

8 functional properties vs the pea study; pea lands ≈ ₹518/kg protein; our cost stays 'not calculable'

Enter a lab result

Lead 1.6 mg/kg (limit 1.0) → NO-GO, before and after; 0.4 keeps TEST-FIRST

Real screenshots from the running prototype (engine 0.8.0), not mock-ups. Live demo on the team laptop; stamped offline snapshots if Wi-Fi fails. Demo video: [attach the ≤ 2-minute video here].

Data and Assumptions

Source	Label
Team final oilseed-cake review (composition, treatments, functionality)	Published · 30 values, 40 rows
Qi et al. 2025, pea protein isolate (80.2% protein)	Published · reference
Murugeswari 2023, aflatoxin B1 survey (118 groundnut-cake samples)	Published
Pea isolate prices (Tridge, IndiaMART), USD/INR, HS 3504 tariff	Published · ranges
Project contaminant limits (7 analytes)	Assumed · internal settings
Freight, duty mode and price scenarios	Assumed · scenario switches
Batch and lab results entered by the user	Measured · none in presets yet
Demo and test values	Synthetic · cannot clear a gate
Organiser datasets: FAOSTAT, IFCT 2017, USDA FDC, FooDB, Open Food Facts, AGMARKNET, data.gov.in	Listed · no values drawn yet

Key results

- Sunflower → Protein Rich Flour: TEST-FIRST, 9 candidate routes
- Pea isolate landed ≈ ₹518/kg protein (₹431–619)
- Lead 1.6 mg/kg → NO-GO

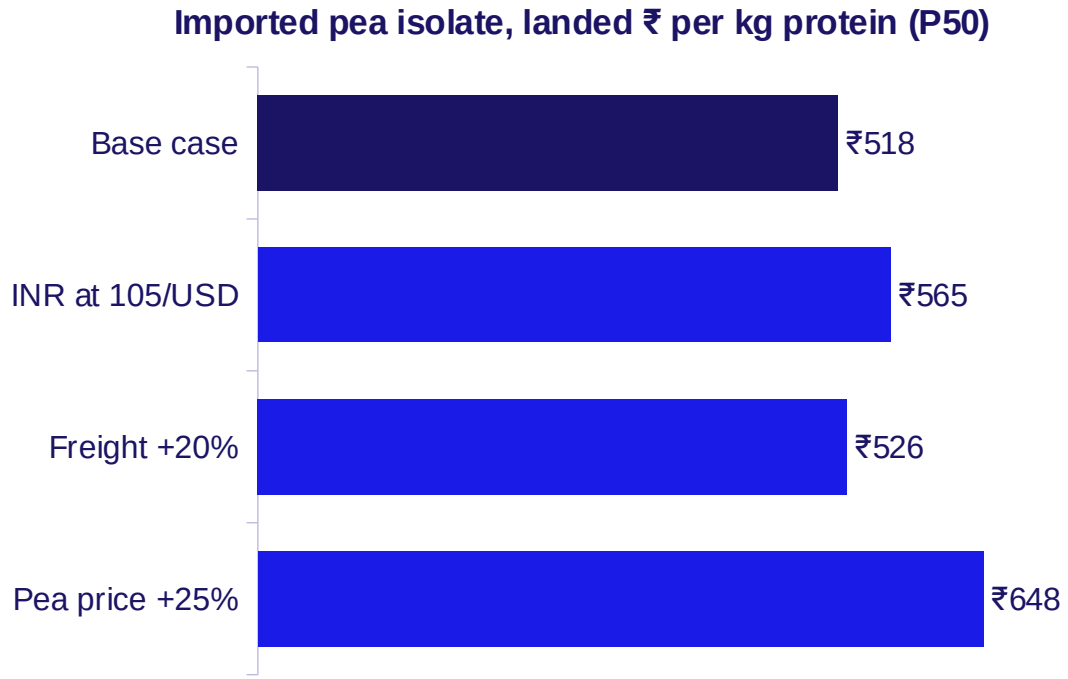
Limits of our data

- Ledger: 161 missing, 41 assumed, 37 to verify
- Many values are review-reported (secondary)
- PhD proximate report withheld pending permission

Error and uncertainty

- Inputs carry P10–P90 ranges
- P(GO), P(spec) and value of information withheld until calibrated
- Missing yield → cost 'not calculable', never guessed

Projected Impact



The bar a local oilseed protein must beat. Scenario model (tagged assumed, grade D) from Tridge prices, USD/INR and the HS 3504 duty.

1. **Aims to replace**
   Part of the ~90% imported plant protein in Indian products, starting with flour- and concentrate-tier uses

2. **Optimises**
   The testing order: safety assays first, so failing lots are diverted before money goes into trials

3. **Market**
   Plant-protein ingredient market expected to grow from ₹4,500 to ₹9,000 crore in five years (PBFIA)

4. **No new harm**
   Only measured values clear safety gates; failed lots stay in the feed chain

Projection, not a result: no cost saving is claimed until yield and process costs are measured in the pilot.

Limitations and Next Steps

What the prototype cannot do yet

- Predict yield, protein recovery or our own ₹/kg protein: no measured process data yet
- Give calibrated probabilities (P(GO), P(spec)) or value of information
- Approve food safety or regulation: limits are project settings to be reviewed
- Draw on the organiser datasets yet; Hindi and voice need real-user review

If the model is wrong

- Safety gates accept only measured values; assumed or synthetic data cannot clear them
- The LLM cannot change numbers; a fixed template is the fallback

The first test we will run

3–5 lots of sunflower cake from food-grade mills:

- Aflatoxin B1 and total aflatoxins (HPLC)
- Lead, cadmium, arsenic and mercury (ICP-MS)
- Hexane residue, if any solvent route was used
- Measured ingredient yield and protein recovery

Then check how often the measurements fall inside the engine's predicted ranges (calibration coverage).

What we need: a food-grade oil mill, an accredited (NABL) lab and a brand to test samples.

References (1/2)

Team documents (unpublished; cited, not shared)

- Team review, cold-pressed oilseed cakes (final draft) [P · secondary]
- Team research plan, 5 Aug 2026 [P]
- Team research evidence matrix, 25 Oct 2026 (75 records) [cited]
- Project contaminant acceptance limits, 24 Oct 2026 [A]
- R. O. Mathur, PhD progress report (LoR, proximate) [withheld]

Papers and reviews

- Qi et al. 2025, functional properties of pea protein isolate [P]
- Murugeswari 2023, aflatoxin B1 in Indian feed materials [P]
- Singh et al. 2022, protein from oilseed cakes [context]
- Detoxification of groundnut meals, Nature 1966 [context]
- El-Kady & Farghaly, aflatoxin inactivation by heat [context]

Market, trade and prices

- PBFIA via Agron: ~90% imported; ₹4,500 → 9,000 crore [P]
- GFI India orphan-crops whitepaper (vegconomist) [context]
- Tridge: pea protein export prices, China [P]
- USD/INR quotes (banklive.net) [P]
- India tariff HS 35040099 (eximpe.com) [P]
- IndiaMART pea protein isolate listings [P]
- IndiaMART oilseed-cake trader listing [A]
- EU provisional anti-dumping duties, Chinese pea protein [cited]
- Canada CBSA pea protein measures [context]
- Expana Plant-Based Protein Report, March 2026 [context]

Labels

- P published · A assumed · M measured · S synthetic. Organiser datasets: next slide.

References (2/2)

Organiser datasets (listed; no values drawn yet)

- FAOSTAT
- IFCT 2017 via @nodef/ifct2017
- USDA FoodData Central
- USDA FoodData Central downloads
- FooDB
- Open Food Facts
- AGMARKNET
- data.gov.in: mandi daily prices
- data.gov.in: variety-wise daily prices

Code and data

- Upcycle Compass repository [make public first]
- Open data pack: linked in the app footer

AI tools

- Amazon Bedrock, model [fill in BEDROCK_MODEL_ID] [optional LLM]
- Groq [optional alternative]
- Claude Code (Anthropic), AI coding assistant [built the code]
- Web Speech API (browser speech input)
- BHASHINI [not used]

Software

- FastAPI, Starlette, Uvicorn, Pydantic
- NumPy, SciPy, pandas
- boto3 (AWS SDK for Python)
- React 18, TypeScript, Vite
- Tailwind CSS
- TanStack Query, Zustand, React Hook Form, Zod
- Radix UI, i18next
- pytest, Hypothesis, Vitest, Playwright, axe-core

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Thank You

